

1 Introduction

A central question in empirical asset pricing is which risk factors actually matter for explaining stock returns. The Fama–French factor models propose that a small set of systematic risk factors—such as market, size, and value—help explain why some portfolios earn higher average returns than others. In practice, however, researchers and practitioners often work with many candidate factors, which raises a natural concern about multiple testing and overfitting: if we run enough regressions, some factors will look “significant” just by chance.

In this project, I study the question:

Which standard Fama–French factors appear to be important for US stock returns once we account for multiple testing and evaluate models based on out-of-sample prediction?

I use data from the Kenneth R. French Data Library: monthly returns on the 25 portfolios formed on size and book-to-market, together with the Fama–French five factors (market, size, value, profitability, and investment) and the risk-free rate. I work with excess returns and treat next month’s portfolio excess return as the outcome, using current factor values as predictors.

Methodologically, I combine standard multiple linear regression with tools from statistical learning covered in this course. First, I fit separate linear regressions for each portfolio and examine factor significance both before and after applying multiple testing corrections (Bonferroni and false discovery rate). Second, I evaluate predictive performance using a time-based train/test split and compare a pooled linear regression model to a LASSO model trained with cross-validation.

The main qualitative finding is that the Fama–French factors have very limited predictive power for next-month excess returns in this setting. Many factor coefficients appear significant at the 5% level when considered in isolation, but once I correct for multiple testing, almost all of this apparent significance disappears. From a predictive perspective, a LASSO model that is free to shrink coefficients towards zero ends up using essentially only an intercept, and its out-of-sample performance is nearly identical to that of the full linear factor model.

2 Background and Data

2.1 Background: factor models and multiple testing

A central idea in empirical asset pricing is that differences in average returns across assets are partly explained by exposure to a small number of systematic risk factors. The Fama–French factor models extend the classical CAPM by adding additional factors beyond the market excess return. In the five-factor version, the excess return on an asset or portfolio at time t is modeled as

$$R_{i,t}^{\text{excess}} = \alpha_i + \beta_{i,M} \text{MktRF}_t + \beta_{i,S} \text{SMB}_t + \beta_{i,H} \text{HML}_t + \beta_{i,R} \text{RMW}_t + \beta_{i,C} \text{CMA}_t + \varepsilon_{i,t},$$

where:

- MktRF_t is the market excess return (market minus risk-free),
- SMB_t is “small minus big,” a size factor,
- HML_t is “high minus low,” a value factor,
- RMW_t is “robust minus weak,” a profitability factor,
- CMA_t is “conservative minus aggressive,” an investment factor.

The slope coefficients $\beta_{i,\cdot}$ are interpreted as factor loadings: for example, a portfolio with a large positive HML loading is viewed as having strong exposure to value risk. This framework is widely used to evaluate mutual funds, decompose portfolio performance, and study cross-sectional patterns in average returns.

Modern empirical work often analyzes many assets and many candidate factors at the same time. If we estimate separate regressions for many portfolios and test the significance of every factor loading, we quickly accumulate a large number of hypothesis tests. Even if all true effects were zero, a conventional 5% significance level would produce “significant” results in about 5% of tests simply by chance. This is the multiple testing problem: without adjustment, we can easily overstate how many factors are truly important.

In parallel, there is a predictive perspective: instead of asking whether a factor loading is statistically different from zero in-sample, we can ask whether including the factors actually improves out-of-sample forecasts of future returns. This connects naturally to the statistical learning tools from class, such as model selection criteria, cross-validation, and regularization methods like the LASSO. In this project, I use both perspectives:

- an inference lens, using multiple testing adjustments to see which factor loadings remain significant across many portfolios; and
- a prediction lens, comparing out-of-sample performance of linear factor models to a LASSO model.

2.2 Data sources

All data come from the Kenneth R. French Data Library, a standard source in academic finance. I combine two datasets:

1. Fama–French Five Factors (2×3), monthly

This dataset provides monthly observations on:

- the market excess return (Mkt–RF),
- SMB (size factor),
- HML (value factor),
- RMW (profitability factor),
- CMA (investment factor),
- and the risk-free rate RF (one-month Treasury bill rate).

The series are reported in percentage form and start in July 1963.

2. 25 Portfolios Formed on Size and Book-to-Market

This dataset reports monthly value-weighted returns for 25 portfolios formed by independently sorting US stocks into:

- 5 size groups (from “small” to “big”), and
- 5 book-to-market (value) groups (from “low B/M” to “high B/M”).

The result is a 5×5 grid of portfolios, such as “SMALL LoBM” (small size, low book-to-market) and “BIG HiBM” (big size, high book-to-market). These portfolios are rebalanced annually and are also reported as monthly percentage returns. They are a canonical test asset set in empirical asset pricing and are often used to assess how well factor models explain differences in returns.

I restrict attention to the period where both datasets overlap. The five-factor series start in July 1963, so I align the 25 portfolio returns to that same start date and use data through the last month available in both series.

2.3 Construction of the analysis dataset

The raw files from the French library require several preprocessing steps before they can be used in the models.

1. Parsing the raw CSV files

The French CSVs include header lines and explanatory notes, followed by the data tables. I read the files with `readr::read_csv`, skipping the non-data lines and retaining only:

- the monthly five-factor panel, and
- the monthly value-weighted returns for the 25 size–value portfolios.

The date dimension is stored as a YYYYMM code (e.g., 196307 for July 1963). I convert this to a Date variable by extracting the year and month and constructing the first day of each month.

2. Converting returns and handling missing codes

Both factors and portfolio returns are originally in percentages. I convert all returns to decimal form by dividing by 100 (so a 2% return becomes 0.02). The documentation indicates that missing returns are coded as large negative values (–99.99 or –999). I map these codes to NA so they do not distort the analysis.

3. Merging factors and portfolios

After cleaning each dataset, I merge them by month. For each month t , the merged dataset contains:

- the five Fama–French factors and the risk-free rate RF , and
- the monthly returns on each of the 25 size–value portfolios.

Because the five-factor data begin in July 1963, the merged panel starts in 1963-07 and runs through the most recent common month. In code, this wide-format panel is stored as `data_model_wide`.

4. Excess returns and forecasting target

I focus on excess returns, which is standard when working with factor models. For each portfolio i and month t , I define the excess return as

$$R_{i,t}^{\text{excess}} = R_{i,t} - RF_t,$$

where $R_{i,t}$ is the portfolio's value-weighted return and RF_t is the one-month risk-free rate, both in decimal form.

Since the goal is forecasting, the dependent variable in all models is the next month's excess return, $R_{i,t+1}^{\text{excess}}$, while the predictors are the current month's factors. For each (portfolio, month) pair, the dataset includes:

- the current month's factors ($\text{MktRF}_t, \text{SMB}_t, \text{HML}_t, \text{RMW}_t, \text{CMA}_t$),
- the current month's excess return $R_{i,t}^{\text{excess}}$, and
- the next month's excess return $R_{i,t+1}^{\text{excess}}$ as the prediction target.

5. Long-format panel structure

For modeling, I reshape the merged data into “long” panel form. Each row corresponds to one portfolio in one month and contains:

- date: month identifier,
- portfolio: a label such as “SMALL LoBM” or “BIG HiBM”,
- excess_return: current month's excess return,
- excess_return_next: next month's excess return,
- Mkt_RF, SMB, HML, RMW, CMA, and RF.

I drop the last observation for each portfolio, since there is no “next month” to predict. The resulting dataset (data_model_long) contains several decades of monthly observations for each of the 25 portfolios, with a consistent set of predictors and outcomes for every period.

2.4 Time-series structure and train–test split

Because these data are time series, a random split of rows into training and test sets would break the time ordering and allow information from the future to leak into the training data. Instead, I construct a time-based split:

- I sort all months chronologically and use the earliest 80% of months as the training period. In this sample, this corresponds to data from July 1963 up to April 2013.
- The remaining 20% of months, from May 2013 onward, form the test period.

All models are fit using only the training period, and the final performance is evaluated on the held-out test period. This setup respects the time structure and aligns with the time-series cross-validation ideas discussed in class.

To visualize the behavior of the target and illustrate the split, Figure 1 plots next-month excess returns over time for two representative portfolios, SMALL LoBM and BIG HiBM. For readability, the figure trims months with extremely large positive or negative returns

(outside $\pm 20\%$) but leaves the vast majority of months intact. The plot shows that next-month excess returns are highly volatile and frequently change sign, with no obvious low-frequency trend. The vertical dashed line marks the 2013-04 cutoff between the training and test periods. This figure highlights both the noisy nature of the forecasting target and the fact that, even over long spans of time, most monthly excess returns are relatively small in magnitude.



Figure 1. Next-month excess returns for SMALL LoBM and BIG HiBM portfolios over time. The y-axis is trimmed to $\pm 20\%$ to improve readability; extreme months are omitted from the plot but retained in the modeling. The vertical dashed line indicates the split between the training period (left) and the test period (right).

3 Modeling Approach

This section explains how I use the Fama–French factors to model and forecast next-month excess returns on the 25 size–value portfolios. The analysis has three main pieces:

1. Start with a simple linear factor regression for a single portfolio, to build intuition.
2. Fit the same regression to all 25 portfolios and examine factor loadings and multiple testing.

3. Pool all portfolios together and compare a standard linear model to a LASSO model focused on out-of-sample prediction.

In all cases, the response is the next month's excess return for a given portfolio, and the predictors are the current month's factor values (MktRF, SMB, HML, RMW, CMA). All models are trained on the early 80% of the sample (July 1963–April 2013) and evaluated on the final 20% (May 2013 onward), as described in the data section and illustrated by the time-series plot in Figure 1.

3.1 Single-portfolio linear factor model

As a starting point, I focus on one representative portfolio: SMALL LoBM (small size, low book-to-market). For this portfolio, I estimate a simple linear regression that uses this month's factors to predict next month's excess return:

$$\text{ExcessReturn}_{t+1} = \beta_0 + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_R \text{RMW}_t + \beta_C \text{CMA}_t + \varepsilon_{t+1}.$$

Before looking at the regression output, I first check the raw relationships between the factors and the outcome. Figure 2 plots next-month excess returns for SMALL LoBM against each factor in the training sample, along with fitted least-squares lines.

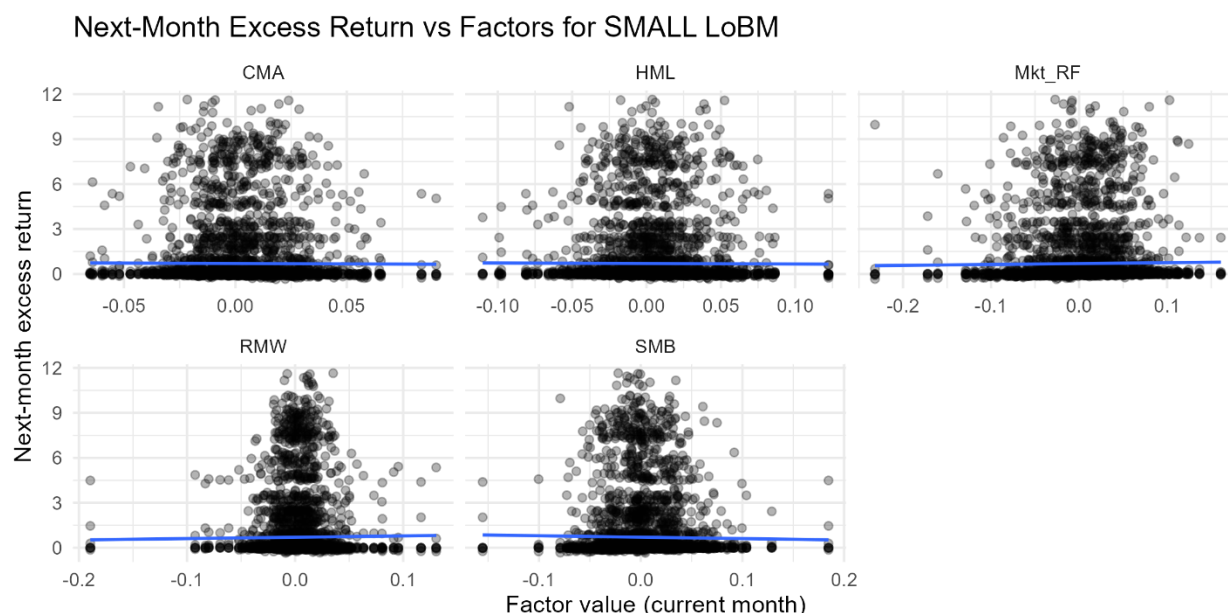


Figure 2. Scatterplots of next-month excess returns for the SMALL LoBM portfolio against each Fama–French factor in the training period, with fitted linear trends.

The panels in Figure 2 show very noisy clouds of points centered near zero, and the fitted lines are almost flat in every case. Visually, there is no strong linear relationship between any single factor and next-month excess return at the monthly horizon. When I fit the full regression, this shows up as very small R-squared values and slope estimates that are not statistically significant. The single-portfolio model is mainly a diagnostic: it highlights how volatile the target is and how limited the predictive content of the factors appears to be for one-month-ahead returns.

3.2 Cross-portfolio regressions and multiple testing

Next, I repeat the same regression for all 25 portfolios. For each portfolio i , I regress its next-month excess return on the five factors, using only the training period:

$$\begin{aligned} \text{ExcessReturn}_{i,t+1} \\ = \beta_{0,i} + \beta_{M,i} \text{MktRF}_t + \beta_{S,i} \text{SMB}_t + \beta_{H,i} \text{HML}_t + \beta_{R,i} \text{RMW}_t + \beta_{C,i} \text{CMA}_t + \varepsilon_{i,t+1}. \end{aligned}$$

This gives 25 separate regressions, 5 slope coefficients per regression, for a total of 125 factor loadings. For each coefficient I record the estimate and its p-value.

Rather than reading 25 tables, I summarize the pattern of factor exposures across portfolios. Figure 3 shows boxplots of the estimated coefficients for each factor across the 25 portfolios.

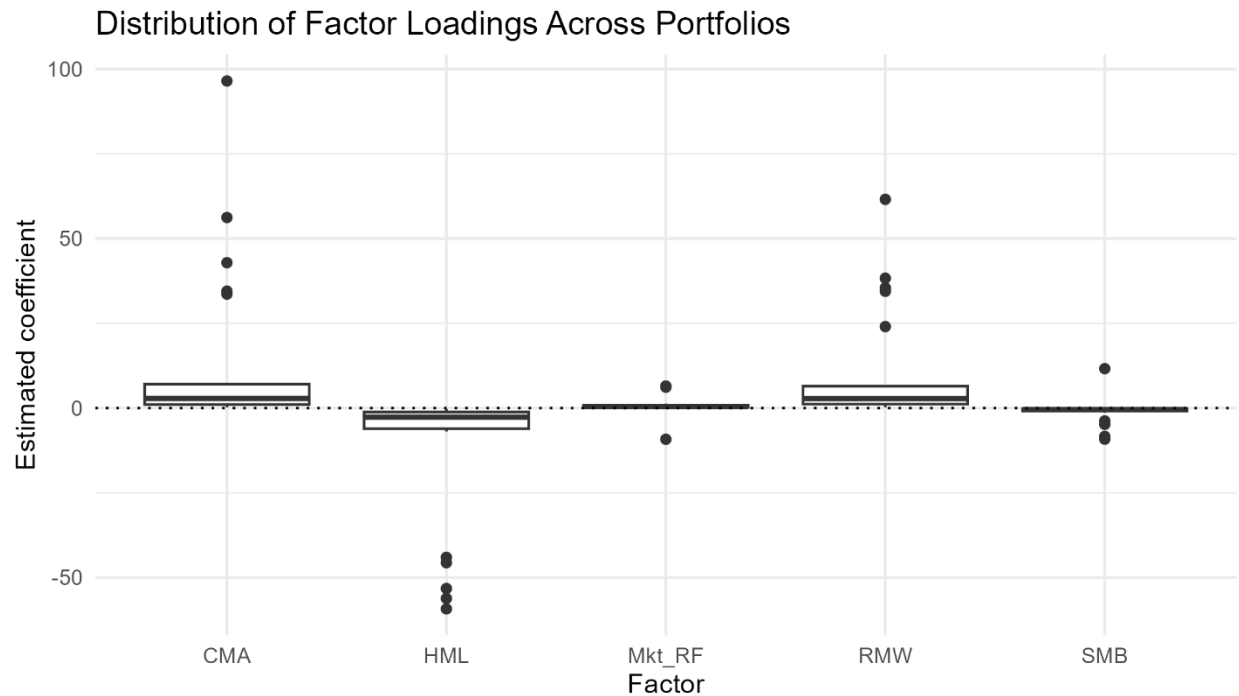


Figure 3. Distribution of estimated factor loadings across the 25 size–value portfolios for each Fama–French factor (training period only).

In Figure 3, the boxes for each factor are generally centered close to zero, with some spread and a few outliers (for example, more extreme HML, RMW, or CMA loadings). There is no factor with a large, consistently positive or negative coefficient across most portfolios. This supports the idea that, in these one-month-ahead regressions, factor exposures are weak and unstable.

Because there are 125 coefficients, there is also a multiple testing issue. If I test each coefficient at the 5% level, I would expect a handful of “significant” results just by chance, even if the true effects were all zero. To address this, I apply two standard corrections:

- a Bonferroni adjustment, which is very conservative and controls the chance of making any false discovery across all 125 tests, and
- a False Discovery Rate (FDR) adjustment (Benjamini–Hochberg), which controls the expected proportion of false positives among the rejected nulls.

For each factor, I count how many of its 25 coefficients are significant at the 5% level under three rules: unadjusted p-values, Bonferroni-adjusted p-values, and FDR-adjusted p-values. In this dataset, the result is stark: none of the 125 coefficients are significant at the 5% level even before adjustment, so the Bonferroni and FDR procedures also identify zero significant loadings for every factor. In other words, across these 25 portfolios, the standard

Fama–French factors do not show any statistically robust one-month-ahead predictive effect once we look at all portfolios jointly and account for multiple comparisons.

3.3 Pooled prediction models: OLS and LASSO

The cross-portfolio regressions focus on the significance of individual coefficients. A complementary question is whether the factors improve out-of-sample prediction when we pool all portfolios together.

To study this, I construct a pooled model where each observation is one (portfolio, month) pair from the panel. The predictors are the same five factors, and the response is next-month excess return. The pooled linear model is:

$$\text{ExcessReturn}_{i,t+1} = \beta_0 + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_R \text{RMW}_t + \beta_C \text{CMA}_t + u_{i,t+1},$$

where the slope coefficients are shared across all portfolios rather than varying by portfolio. This model ignores portfolio-specific intercepts and simply asks whether a single set of factor coefficients can improve forecasts on average, across all 25 portfolios.

I first estimate this pooled model with ordinary least squares (OLS) on the training data, then compute predictions for the test period and summarize performance using the root mean squared error (RMSE) of the forecast errors.

To bring in a regularization method from class, I also fit a LASSO model on the same pooled design matrix. The LASSO adds an ℓ_1 penalty to the regression and can shrink some coefficients all the way to zero, automatically performing variable selection. I select the LASSO penalty parameter λ using 10-fold cross-validation on the training sample. In this application, the cross-validated mean squared error is essentially flat across λ : the error only varies by a tiny amount as λ increases. Under the usual “one-standard-error” rule, this flat curve leads to a relatively large λ , and the LASSO shrinks all five factor slopes to zero, leaving only an intercept term.

In other words, the LASSO’s best strategy is to drop all factor predictors and simply predict the overall mean next-month excess return. When I compare test-set RMSE for the pooled OLS model and this intercept-only LASSO model, they are almost identical. This reinforces the message from the scatterplots and coefficient distributions: at the monthly horizon and in this setup, the Fama–French factors do not provide meaningful incremental predictive power for next-month portfolio excess returns, either from a coefficient-significance perspective or from an out-of-sample prediction perspective.

4 Results

This section summarizes what the models find about the relationship between Fama–French factors and next-month excess returns on the 25 size–value portfolios. I first discuss the single-portfolio regression for SMALL LoBM, then the cross-portfolio regressions and multiple testing, and finally the pooled prediction results comparing OLS and LASSO.

4.1 Single-portfolio regression: SMALL LoBM

For the SMALL LoBM portfolio, I regress next-month excess returns on the five Fama–French factors using only the training period (July 1963–April 2013). The regression includes an intercept and the factors MktRF, SMB, HML, RMW, and CMA.

Consistent with the scatterplots in Figure 2, the regression finds very weak explanatory power:

- All five factor slope estimates are small in magnitude and statistically insignificant at conventional levels.
- The R-squared is essentially zero (around 0.001), meaning the factors explain almost none of the month-to-month variation in next-month excess returns for this portfolio.
- The residual standard error is large relative to the typical monthly excess return, which reflects the high volatility and occasional extreme outliers in the series.

To evaluate predictive performance, I use the fitted model to forecast next-month excess returns for SMALL LoBM in the test period (May 2013 onward). The test-set RMSE for this single-portfolio OLS model is about 1.05 (in decimal units). This is quite large compared to the typical magnitude of monthly excess returns, and it confirms what Figure 2 suggests: at the one-month horizon, the linear factor model has almost no ability to forecast next-month excess returns for this portfolio.

4.2 Cross-portfolio regressions and multiple testing

I then repeat the same regression for all 25 portfolios, still using only the training period. This yields:

- 25 regressions,
- 5 factor slopes per regression,
- for a total of 125 factor loadings.

Figure 3 summarizes these estimated coefficients across portfolios by plotting, for each factor, the distribution of its 25 loadings. Several patterns emerge:

- For every factor, the median coefficient is close to zero, and the interquartile range is relatively narrow.
- There are some outliers for HML, RMW, and CMA, but these are exceptions rather than a consistent pattern across portfolios.
- No factor shows large, systematically positive or negative coefficients across the cross-section.

To assess statistical significance, I examine the p-values for all 125 factor loadings. Even before any adjustment, none of the coefficients are significant at the 5% level. Applying Bonferroni and FDR corrections does not change this: all three significance counts (unadjusted, Bonferroni, and FDR) are zero for every factor.

Taken together, the boxplots and the multiple testing results paint a consistent picture. From an inference perspective:

- At the monthly horizon and in this setup, the standard Fama–French factors do not display any robust, statistically significant effect on next-month excess returns across the 25 size–value portfolios.
- Any apparent nonzero coefficients are small, noisy, and not distinguishable from zero once we account for sampling variation and the large number of tests.

4.3 Pooled prediction: OLS versus LASSO

The cross-portfolio regressions focus on individual coefficients. To evaluate predictive performance more directly, I next pool all portfolios together and fit models where each observation is a (portfolio, month) pair with:

- predictors: the current month's MktRF, SMB, HML, RMW, and CMA, and
- outcome: next-month excess return for that portfolio.

I estimate two pooled models on the training sample:

1. Pooled OLS: a standard linear regression with a single set of slopes shared across all portfolios.
2. Pooled LASSO: a regularized regression that penalizes the absolute size of the coefficients and can shrink some to zero.

For the LASSO, I choose the penalty parameter λ using 10-fold cross-validation. In this application, the cross-validated mean squared error is essentially flat across λ (it only changes by a tiny amount over the whole grid), indicating that the amount of shrinkage has almost no impact on average prediction error. Under the usual one-standard-error rule, the selected λ is relatively large, and the LASSO shrinks all five factor slopes to zero, leaving only an intercept.

I then compute test-set predictions from both models and summarize performance using the RMSE on the held-out period. At a high level, the results look like this:

- The pooled OLS model and the intercept-only LASSO model have almost identical test RMSE (both around 123.6 in squared-error units on the pooled scale).
- The LASSO's best strategy is essentially to ignore all factor information and predict a constant mean return; adding factors does not improve out-of-sample accuracy.

These pooled results are consistent with the earlier findings:

- For a single portfolio (SMALL LoBM), factor-based forecasts are extremely noisy and add little value.
- Across 25 portfolios, estimated factor loadings are clustered near zero and never statistically significant.
- When we give the LASSO freedom to decide whether to use the factors at all, it chooses to shrink every slope to zero and achieves the same predictive accuracy as the more complex pooled OLS model.

Overall, the results suggest that, in this particular setting—monthly data, these 25 size–value portfolios, and the standard set of Fama–French factors—the factors have very limited predictive power for next-month excess returns. Both from a multiple-testing perspective and from an out-of-sample prediction perspective, the data are consistent with the idea that one-month-ahead excess returns are dominated by noise rather than by persistent factor-driven predictability.

5 Discussion

5.1 Main findings

The goal of this project was to ask whether the standard Fama–French factors help explain or predict next-month excess returns on the 25 size–value portfolios, once we use tools from class like multiple testing adjustments and regularization.

Across all analyses, the answer is consistent:

- For a single portfolio (SMALL LoBM), factor regressions have near-zero R^2 and insignificant slopes.
- Across all 25 portfolios, the 125 factor loadings are centered close to zero, and none are significant at the 5% level, even before Bonferroni or FDR corrections.
- In pooled prediction, the LASSO chooses to set all factor slopes to zero, and its test RMSE is essentially identical to that of pooled OLS.

Taken together, this suggests that, in this setup, the five Fama–French factors provide very little useful information for one-month-ahead excess return prediction.

5.2 Interpretation and limitations

These results do *not* contradict the large literature showing that Fama–French factors help explain average returns over long horizons. Instead, they suggest that at a short, monthly horizon, factor signals are very small relative to the noise in returns, at least for these 25 diversified portfolios and this simple linear forecasting setup.

There are several limitations:

- The horizon is fixed at one month; factors may matter more for longer horizons.
- Models are linear and low-dimensional, with no interactions or nonlinear terms.
- Time-series structure beyond a rolling train/test split (e.g., AR terms, volatility dynamics) is not modeled explicitly.
- The analysis is restricted to size–value portfolios; other test assets might behave differently.

Given these constraints, the conclusion is specific: with this data, horizon, and model class, factor-based forecasts are hard to distinguish from simply predicting a constant mean.

5.3 Possible extensions

Natural extensions would be to:

- Try longer horizons (e.g., multi-month or annual returns).
- Use different test assets (momentum-sorted or industry portfolios).
- Compare to more flexible methods from class (nonlinear models or tree-based methods) while still respecting time ordering.

- Expand the set of predictors to a larger “factor zoo” and apply multiple testing and model selection in that richer setting.

Overall, the project shows how regression, multiple testing, and regularization can work together: here, they all point to the same simple message—at the monthly level, these standard factors do not deliver robust, out-of-sample predictability for the 25 size–value portfolios.

References

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